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Unveiling the molecular basis of *Pleurocybella porrigens* lectin (PPL)–glycan recognition by NMR and X-ray crystallography

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ABSTRACT

Lectins of the R-type superfamily recognize specific glycan motifs through β-trefoil carbohydrate-binding domains and play central roles in cell communication, immunity, and host–pathogen interactions. Here, we investigated the incompletely characterized glycan-binding specificity of *Pleurocybella porrigens* lectin (PPL) through an integrated structural approach combining NMR spectroscopy and X-ray crystallography. Three structurally related ligands of increasing biological complexity were analyzed: *N*-acetylgalactosamine (GalNAc), galactose (Gal), and galacto-N-biose (Galβ1 → 3GalNAc), a minimal mucin-type O-glycan epitope associated with cancer-related glycosylation. STD NMR experiments defined the ligand-binding epitopes in solution and revealed different contributions of individual sugar protons to PPL recognition. GalNAc displayed an extensive interaction profile, with the *N*-acetyl substituent playing a key role in stabilizing binding; whereas, in the case of Gal, PPL showed an exquisite preference for the recognition of the β-anomer. X-ray crystal structures, obtained by soaking and co-crystallization experiments, provided atomic-resolution insight into ligand accommodation within the β-trefoil binding pocket, identifying hydrogen-bonding networks and apolar interactions responsible for glycan recognition. The combined NMR and crystallographic data establish consistent models for PPL binding to various ligands. Our findings clarify the structural basis of Gal/GalNAc selectivity in PPL and support the broader utility of ricin-B–like lectins as probes for biologically and clinically relevant O-glycan epitopes. More generally, this work highlights the complementarity of NMR and X-ray crystallography for the integrated characterization of lectin–glycan interactions.

1. Introduction

Lectins recognize specific carbohydrate motifs on cell surfaces and translate this information into biological functions. Their ability to discriminate subtle differences in glycan stereochemistry and linkages enables them to sense cell state and tissue context. These recognition events underlie essential processes such as adhesion, trafficking, immune signaling, and host–pathogen interactions. Altered glycosylation patterns often reflect disease and can be exploited for detection and therapeutic targeting [1–5].

Pleurocybella porrigens lectin (PPL) originates from the basidiomycete *Pleurocybella porrigens*, commonly known as the “angel's wing” mushroom. For decades, it was considered edible in Japan. In autumn 2004, however, a cluster of acute encephalopathy and renal impairment was associated with its consumption [6]. Following this event, health authorities classified the mushroom as poisonous. Subsequent investigations identified several components of a toxin which can disrupt the blood–brain barrier and is lethal to rats. The specific contribution of each component to toxicity remains under study, but the lectin component, called PPL, is believed to be responsible for attachment of the toxin complex to target cells [7–9].

PPL belongs to the R-type lectin superfamily. Members of this family share a β-trefoil fold, first identified in the ricin B chain, with three similar subdomains arranged around a pseudo-threefold axis. This architecture occurs across species and is often combined with additional biological functions, including toxin modules [10,11]. Recent structural

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analyses of PPL reveal a novel hexameric assembly of β-trefoil subunits [12]. Studies of other ricin-B–like fungal lectins highlight their adaptability as molecular probes for cancer-associated glycans, underlining the diagnostic and functional potential of this fold when its specificity is clearly defined [13].

Previous functional studies provided the first clues to PPL molecular preferences. Hemagglutination inhibition assays consistently showed a strong affinity for *N*-acetylgalactosamine (GalNAc), weaker binding to galactose (Gal) and lactose (Lac), and potent inhibition by asialomucins, consistent with recognition of mucin-type O-glycan motifs [14]. Glycoarray analysis [15], using a wider selection of ligands, confirmed that α-GalNAc bound strongly; ITC analysis showed that GalNAc free in solution has an affinity for PPL roughly 40-times stronger than that of galactose [12]. The cancer-associated glycosylation biomarker TF antigen consists of Galβ(1 → 3)-D-GalNAc, O-linked to serine or threonine at the cancer cell surface. TF antigen was not seen to bind PPL in the glycoarray, and attempts to model TF antigen into the binding pocket of PPL by simple rigid superposition of the galactose rings produced steric clashes between the GalNAc moiety and Gly23/Ser24 [12]; however, the biomedical relevance of this ligand could not be discounted. These observations guided the selection of ligands analyzed here, allowing a systematic progression from simple monosaccharides to a minimal O-glycan epitope.

To dissect PPL–glycan recognition, we examined three structurally related ligands. GalNAc was chosen first because it is the preferred sugar in previous studies and initiates mucin-type O-glycosylation, forming the Tn antigen. Gal serves as a close comparator, allowing us to evaluate the role of the *N*-acetyl group. Finally, we investigated the interaction with Galacto-N-biose (D-Galβ1 → 3-D-GalNAc), the disaccharide of the TF antigen [16,17]. Many tumors display truncated mucin-type O-glycans on their surfaces. The simplest unit, the Tn antigen (α-GalNAc-Ser/Thr), and its galactosylated derivative, the core-1/T antigen (Galβ1-3GalNAc-O-Ser/Thr), appear in a wide range of cancers and serve as diagnostic markers because they remain accessible to carbohydrate-binding proteins [18]. Despite the apparent lack of TF antigen binding reported by Adachi et al. [12], we chose to test Galacto-N-biose as NMR allows high sugar concentrations to be used, while glycoarrays may fail to detect very weakly binding ligands with high off-rates [15,19]. A molecular-level understanding of how a Gal/GalNAc-selective fungal lectin recognizes these structural units provides a foundation for glycophenotyping strategies and for the rational design of selective glycan-recognition probes [20–22].

Our approach combines ligand-based NMR spectroscopy with X-ray crystallography on the same protein–ligand complexes. Saturation transfer difference (STD) NMR identifies the ligand epitope in solution by transferring saturation from the protein to the bound ligand. Protons in close contact with the receptor experience stronger saturation and therefore show higher relative intensity in the corresponding difference spectrum. This technique enables rapid and sensitive detection of weak ligand–target interactions in solution without isotopic labeling, while also providing quantitative information about the ligand binding epitope [23–26]. In parallel, X-ray crystallography allows us to study the effect of the sugar binding on the structure of ligand-free protein, and to define atomic contacts, ring orientation, hydrogen-bonding networks, and stacking interactions within the binding site [27].

PPL is particularly well suited to this integrated analysis. The protein is structurally well defined, including its unusual hexameric organization, which provides a stable framework for ligand binding. Thus, here we present a combined analysis of PPL binding to GalNAc, Gal, and Galacto-N-biose, using STD NMR to map the interaction epitopes and compare their relative contributions to binding, and correlating these with structural models provided by X-ray crystallography. The combined dataset yields a cross-validated recognition model that connects solution dynamics with binding-site chemistry, and demonstrates the utility of combined NMR and X-ray crystallography analysis for lectin–glycan systems.

2. Materials and methods

2.1. Protein preparation

PPL was cloned and purified following the procedure previously published [12,28] at the Drug Design Laboratory of Yokohama City University, Yokohama, Japan, and shipped to the Department of Chemical Sciences of University of Naples Federico II, Naples, Italy, for further experiments. Briefly, a codon-optimized gene was inserted into the pET28b plasmid and transformed into *Escherichia coli* BL21(DE3) cells to produce PPL carrying a cleavable N-terminal histidine tag. The pellet was lysed using 100 mM Tris-HCl pH 8.0, 150 mM NaCl. Purification was accomplished using a Ni-NTA column equilibrated with 50 mM Hepes pH 7.5, 100 mM NaCl, followed by size-exclusion chromatography (SEC; HiLoad 16/600 Superdex 200 pg, GE) in 50 mM HEPES pH 7.5, 100 mM NaCl. Purified samples were then lyophilized after desalting.

2.2. NMR spectroscopy

All the NMR experiments were recorded on a Bruker AVANCE NEO 600 MHz equipped with a cryoprobe, at a temperature of 298 K. Data acquisition and processing were performed by using TOPSPIN 4.4 software. Samples were prepared in deuterated phosphate buffered saline (PBS) at pH 7.4, using 2,2,3,3-d(4)-3-(trimethylsilyl)propionic acid sodium salt (TSP, 20 μM) as internal reference for spectra calibration.

The chemical shifts of D-Gal, D-GalNAc and Galacto-N-biose were assigned by a combination of homo- and heteronuclear 2D NMR spectroscopic experiments, including correlation spectroscopy (COSY), total correlations spectroscopy (TOCSY), nuclear Overhauser effect spectroscopy (NOESY) and $^{1}H—^{13}C$ heteronuclear single-quantum coherence (HSQC) experiments.

Ligand-based NMR experiments were conducted with protein: ligand ratios ranging from 1:20 to 1:80. Finally, protein and ligand concentrations were set at 20 μM and 1.6 mM, respectively. Saturation Transfer Difference (STD) NMR experiments were acquired with 64 k data points, and the FIDs were zero-filled to 128 k points before Fourier transformation to improve spectral digitization and peak definition. The protein resonances were selectively irradiated using 40 G pulses with a length of 50 ms, setting the off-resonance pulse frequency at 40 ppm and the on-resonance pulses in aromatic and aliphatic region (7.5 ppm and 0 ppm). A saturation time of 2 s was selected. In order to suppress protein signals, a 20 ms spinlock pulse was applied. The ligand epitope mapping was obtained by the calculation of the ratio of the STD signals ($I_0 - I_{sat}$) and the relative peak intensity of the unsaturated reference spectrum (off-resonance, I_0). The highest STD signal was set to 100% and all the other STD were normalized to this value. Control spectra were also acquired in the absence of protein to verify that the observed STD responses originated from specific protein–ligand interactions.

2.3. Crystallization and X-ray diffraction data collection

PPL was crystallized using the conditions previously reported [12]. Briefly, the protein (10 mg ml^{-1}) was crystallized at 298 K using the hanging drop vapor diffusion method and the reservoir constituted of 0.15 M Tris-HCl pH 9.5, 4.0 M sodium nitrate. Crystals of the ligand-free protein were then soaked with D-Gal, D-GalNAc or Galacto-N-biose to obtain structures of the complexes. Crystals of the complexes with D-GalNAc and Galacto-N-biose only diffracted to low resolution, so the protein was co-crystallized with the same sugars, after incubating for 3 h at room temperature the protein with the ligand using a protein chain to sugar molar ratio of 1:1. These samples were then used to obtain crystals of the complexes, using the same experimental conditions that allowed the crystallization of the ligand-free protein. All the crystals were cryoprotected using a solution formed by the reservoir and 20% glycerol and flash-cooled in liquid nitrogen. Diffraction data were then collected

using cryo-cooled crystals at 100 K at Elettra synchrotron in Trieste and processed with Autoproc [29]. Data collection statistics are summarized in Table S2.

2.4. Structure resolution and refinement

All the structures were solved by molecular replacement using the structure of the complex of PPL with D-GalNAc as a starting model (PDB entry 9KL2 [12],) and the program Phaser [30]. Model building and inspection of the electron density maps were carried out using Coot [31]. The structures were isotropically refined using Refmac [32] from CCP4 [33]. Non-crystallographic symmetry averaging was not applied. The final structural models were deposited in the Protein Data Bank (PDB) under the accession codes 31VV, 31VW, 31WZ, 31WG, 31WE, 31XF. Final refinement statistics are provided in Table S2. All figures were created with Pymol (www.pymol.org).

3. Results & discussion

3.1. NMR spectroscopy

The molecular recognition of GalNAc, Gal and the disaccharide Galacto-N-biose (D-Galβ1 → 3-D-GalNAc) by PPL was first investigated using Saturation Transfer Difference (STD) NMR to define the ligand-binding epitopes. The results clearly showed that PPL recognizes D-GalNAc, the tightest binding ligand, with no anomer preference (α vs β). Several STD enhancements specific to either the α or β-anomer were observed. Ligand epitope mapping was normalized to the $H4_\alpha$ proton, the most intense and well-resolved STD signal, set to 100% and the other intensities were calculated accordingly (Fig. 1). Although some STD peaks could not be assigned and quantified because of resonance overlapping, strong STD effects were observed for H1, H2, and H5 of the α-anomer, with STD% of 64%, 75%, and 88%, respectively, and for H1 and H2 of the β-anomer, with STD% of 70% and 86%, respectively. A minor response from the *N*-acetyl group was also detected.

In the case of D-Gal, surprisingly, PPL exhibited a clear preference for the β-anomer. As shown in Fig. 2, in the presence of reducing sugar, no STD signals were observed for the α-anomer, while several STD enhancements were detected for the β-anomer. In particular, the largest normalized STD responses were observed for the protons H2, H3 and H4 (99–100%) of the β-anomer. Protons H1 and H6 also showed strong contributions to the interaction (93% and 72%, respectively), whereas proton H5 gave a lower response (55%) (Fig. 2).

Further evidence for the protein preference for the β-anomer was provided by STD-NMR measurements acquired on PPL in the presence of α-GalOMe (data not shown), where no significant STD enhancements were detected.

We then moved to Galacto-N-biose. NMR revealed that PPL was able to accommodate the disaccharide in its binding pocket, establishing the main contacts with the β-D-Gal residue. The largest normalized STD response was observed for H4 Gal proton, and significant STD effects were also observed for protons H2 (83%) and H3. STD effects below 40% were detected for protons H5 and H6 of the Gal residue. Overall, the STD results suggested that the terminal Gal was closely associated with the protein binding site, while the GalNAc moiety was oriented farther from the protein surface. Only its *N*-acetyl group gave rise to a clear STD signal (50%) (Fig. 3).

4. X-ray crystallography

4.1. Crystals of PPL in complex with D-GalNAc: high resolution data

The X-ray structure of PPL complexed with D-GalNAc was previously refined to 2.0 Å resolution [12]. Crystals of this complex belong to the

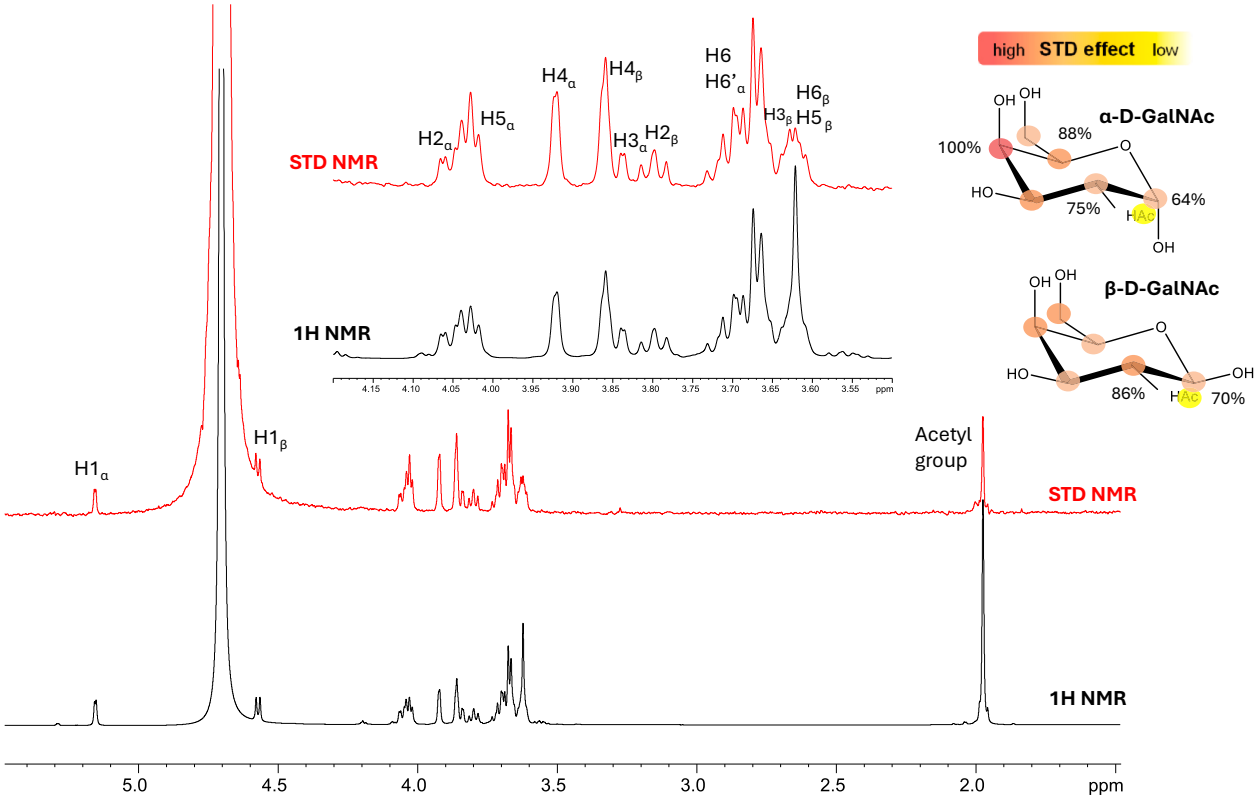


Fig. 1. STD NMR analysis of D-GalNAc bound to PPL. Superimposition of the STD NMR spectrum acquired with on-resonance saturation at 7.5 ppm (red) and the ^{1}H NMR spectrum (black) of PPL: D-GalNAc mixture at a molar ratio of 1:80. An expanded view of the most crowded region has been included for clarity. The percentage of STD effects has been reported only for well resolved resonances.

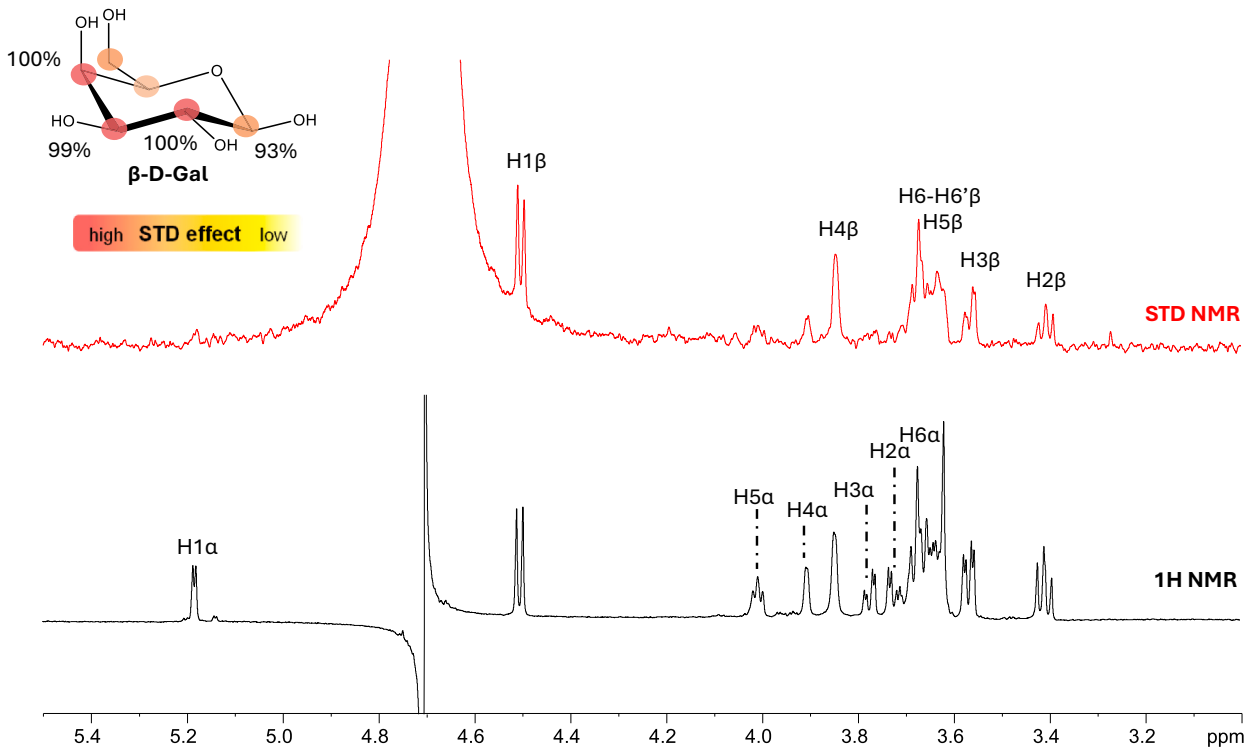


Fig. 2. STD NMR analysis of D-Gal bound to PPL. Superimposition of the STD NMR spectrum acquired with on-resonance saturation at 7.5 ppm (red) and the unsaturated reference spectrum (black) of PPL: D-Gal mixture at a molar ratio of 1:80. STD spectra and ligand epitope mapping highlight only interactions that contribute to the interaction with a STD % above 20%. The resonances labeled in the ^{1}H NMR spectrum correspond to protons that do not contribute to binding. Only those protons showing STD enhancements have been assigned in the STD NMR spectrum.

space group $C222_1$, and contain three protein chains in the asymmetric unit (a.u.) that reconstitute the complete hexameric structure by symmetry (Fig. 4a, PDB code 9KL2). This published model includes residues 1–123/124 and 128–137; GalNAc interacts with residues Asp20 – Gly23 - Ser24 – Asn42 in the three chains of the protein that are present in the a. u. (Table S1 and Fig. 4b-d).

First, we reproduced these crystals and collected X-ray diffraction to high resolution (1.64 Å). The structure of PPL/D-GalNAc derived from the present structural analysis provides a more accurate and better-defined structure of the complex. In particular, the new data allow the reconstruction of the loop 124–127 (Fig. 5a) and of the N-terminal His-tag (Fig. 5b), both unmodeled in the previous structure. In our crystals, PPL binds the D-GalNAc only in two out of the three protein chains of the a.u. (Fig. 6a-c). Inspection of the 2Fo-Fc electron density (e.d.) map suggests that the α-anomer is preferentially bound to the protein. However, a negative e.d. peak is present on the O1 atom of D-GalNAc α-anomer model and a positive e.d. peak is present close to the C1 atom of D-GalNAc α-anomer model in the residual Fo-Fc e.d. map in chain A. This finding suggests that a small proportion of the β-anomer is also present in the crystal, in complete agreement with the previous X-ray structure solved by Adachi et al. [12]

Interactions between the sugar and protein are as previously described: atoms C3, C4 and C5 form van der Waals interactions with atoms of the Trp35 side-chain; atoms O3 and O4 form hydrogen bonds with the side-chain oxygen atoms of Asp20; O3 also forms a bond with the side-chain of Asn42 and O4 bonds with the main-chain N atom of Gly23. The O7 atom of acetyl group interacts with the side chain of Ser24. Table S1 shows a detailed comparison between the interactions of D-GalNAc and PPL observed in previous structures and in the models reported here.

4.2. Crystals of ligand-free PPL

To evaluate possible effects of the binding of D-GalNAc on the structure of the ligand-free PPL, the protein was crystallized in the absence of carbohydrates. Two different crystals of ligand-free PPL were obtained. The first one belongs to the space group $P2_1$, diffracts X-ray to 1.92 Å resolution and contains a protein hexamer in the asymmetric unit. The second one belongs to the space group $P2_12_12$, diffracts X-ray at 1.96 Å resolution and has three protein chains in the a.u.

As expected, in both crystals the structure of ligand-free protein adopts a homohexameric assembly with a total buried surface area of about 7500 Å^2. The PPL monomer adopts the conserved β-trefoil fold. The r.m.s.d. between individual PPL chains is 0.60 Å or lower (excluding the loop residues 124–128), indicating high structural similarity between the individual monomers. Every monomer directly contacts two others, burying between 1000 Å^2 and 1500 Å^2 at each interface, about half of the buried surface being polar.

The D-GalNAc binding sites in the structures of liganded and ligand-free PPL are highly similar: water molecules occupy the positions of the oxygens of D-GalNAc in the $P2_1$ structure and in one of the three chains of the $P2_12_12$ structure (Fig. 7a-f and Fig. 7i), while a nitrate ion is observed in the other two chains of the $P2_12_12$ structure (Fig. 7g-h).

4.3. Crystals of PPL in complex with galactose

To understand the differences of behavior between Gal and GalNAc, crystals of the PPL complex with D-Gal were obtained both by co-crystallization and crystal soaking. Crystals of PPL soaked for 7 days with an excess of D-Gal belong to the space group $P2_12_12$ while those obtained by co-crystallization (using a protein sample incubated 3 h with D-Gal in 1:1 protein monomer to sugar molar ratio) belong to the space group $P2_1$. These crystals are isomorphous to the two crystal forms of the ligand-free protein, allowing a strict comparison of the physico-

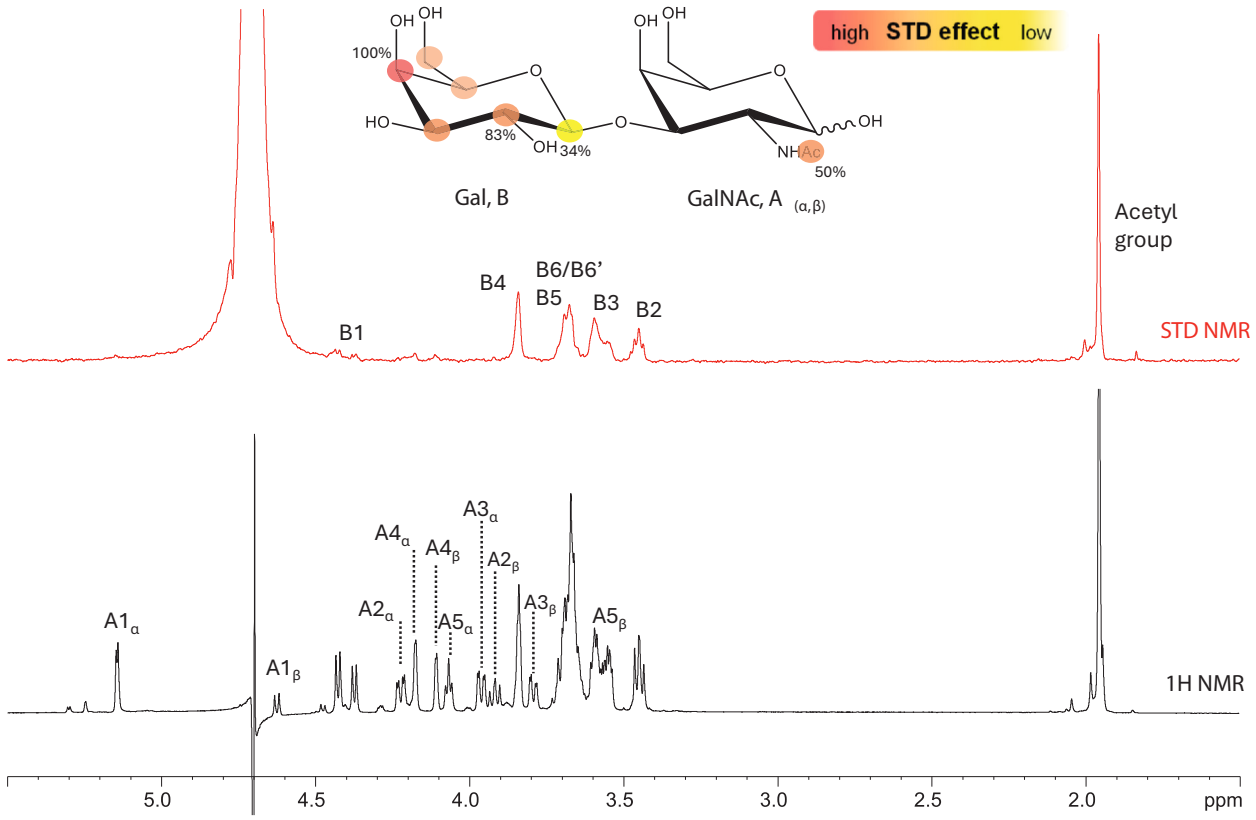


Fig. 3. STD NMR analysis of Galacto-N-biose bound to PPL. Superimposition of the STD NMR spectrum, acquired with on-resonance saturation at 7.5 ppm (red), and the unsaturated reference spectrum (black) of PPL: Galacto-N-biose mixture at a molar ratio of 1:80. STD spectra and ligand epitope mapping highlight only interactions that contribute to the interaction with a STD % above 20%. The resonances labeled in the ^{1}H NMR spectrum correspond to protons that do not contribute to binding. Only those protons showing STD enhancements have been assigned in the STD NMR spectrum.

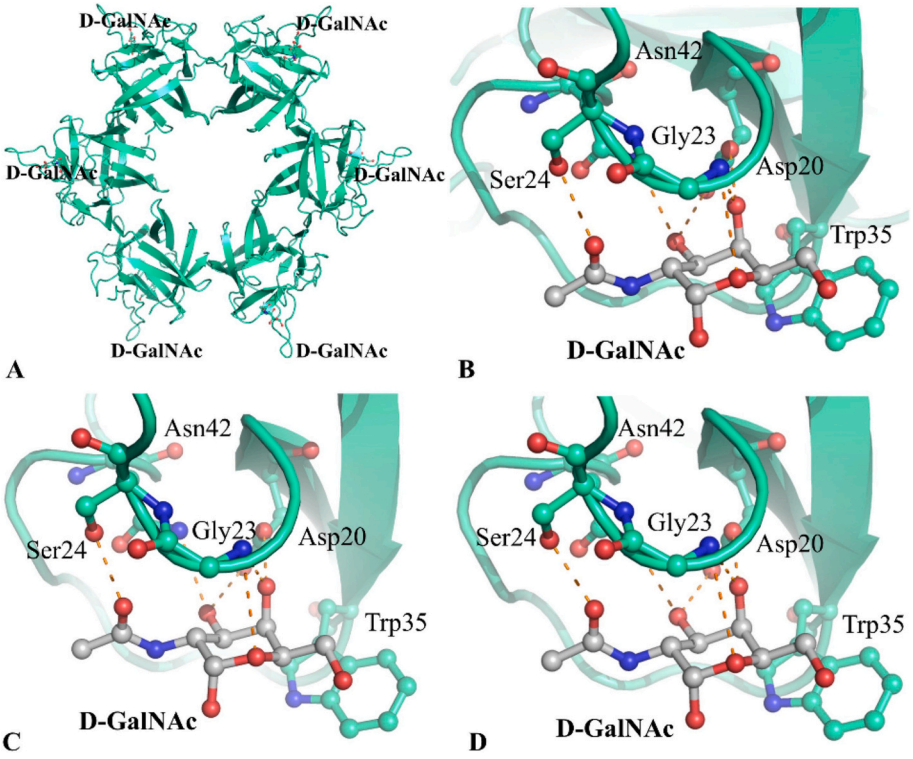


Fig. 4. Overall structure of the complex formed by PPL with D-GalNAc (A) and details of the binding sites of D-GalNAc in the three PPL chains in the asymmetric unit of the crystals solved by Adachi et al. (PDB code 9KL2) [12]. The hydrogen bonds formed with Asp20, Gly23, Ser 24 and Asn42 are shown as orange dotted lines. Ligand carbon atoms are in grey.

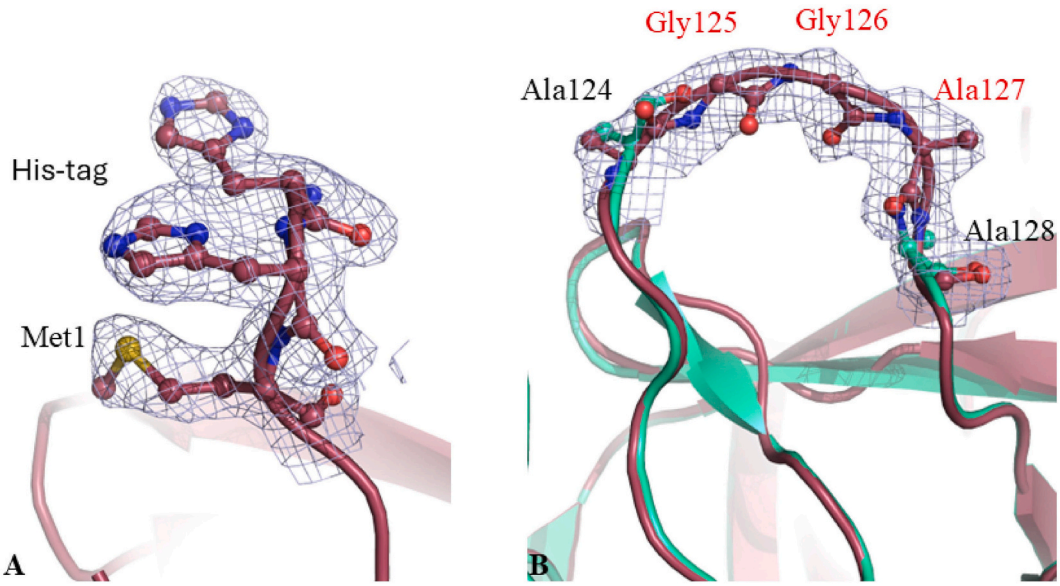


Fig. 5. 2Fo-Fc electron density maps (light blue) contoured at 1.0 σ and Cα trace cartoons (colored raspberry) showing (A) the N-terminus of one chain including residues of the His-tag and (B) the loop formed by residues 124–127 in the structure of PPL with D-GalNAc. In panel B the earlier model (PDB 9KL2, shown in green) was superimposed on the PPL/D-GalNAc described here. Residues absent from the 9KL2 structure are labeled in red.

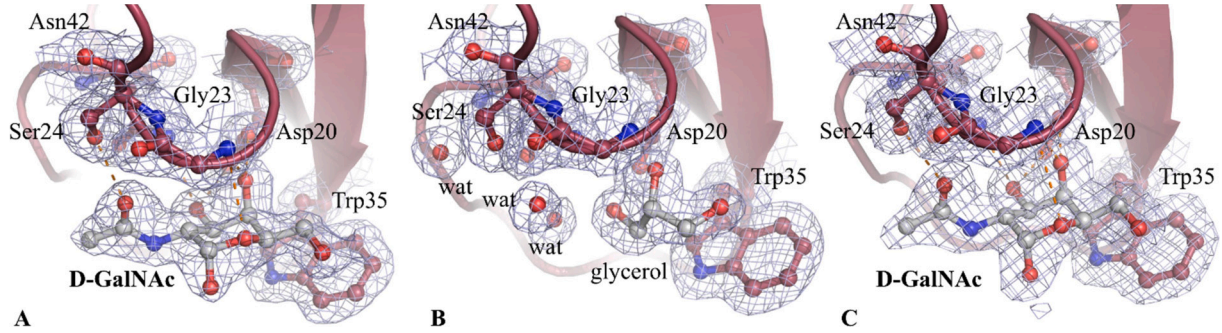


Fig. 6. Electron density maps covering the three ligand binding sites of the a.u. 2Fo-Fc electron density maps are shown in light blue at 1.0 σ. Water molecules are shown as red spheres. The hydrogen bonds formed by sugar ligands with Asp20, Gly23, Ser 24 and Asn42 are shown as orange dotted lines. (A) D-GalNAc, (B) glycerol occupies part of the D-GalNAc binding site, (C) D-GalNAc. Ligand carbon atoms are in grey.

chemical features of the D-Gal binding site in the ligand-free structure and in the complex. In the $P2_1$ structure of the PPL/D-Gal complex the sugar binds five out of the six protein chains in the a.u., while in the other crystal, two of the three PPL chains bind D-Gal. The sugar binding does not alter either the overall structure or the sugar binding site. In agreement with NMR data, only the β-anomer form was observed (Fig. 8). In marked contrast to the X-ray models with GalNAc bound, in almost all the protein subunits of the PPL/D-Gal structures, O2 and O3 of the sugar form hydrogen bonds with the carboxyl oxygens of Asp20, O4 accepts a hydrogen bond from the N atom of Gly23, while O2 atom forms a hydrogen bond with either the side chain of Asn42 or with a water molecule that in turn is bound to the side chain of Ser24 (Fig. 8). The only exception is observed in the chain B of the $P2_12_12$ structure (Fig. 8h). Here, like GalNAc, D-Gal interacts with the side chain of Asp20 through the O3 and O4 atoms, with the side chain of Asn42 through O3 atom and with the O of Leu21 through the O4 atom. The interactions formed by PPL with different ligands are reported in Table S1.

4.4. Structure of the complex of PPL with Galacto-N-biose

To complete the structural characterization of complexes of PPL with representative sugars, the PPL/Galacto-N-biose complex was crystallized in the space group $P2_1$ and diffraction data were collected to 2.79 Å resolution. Galacto-N-biose is anchored to the protein via hydrogen bonding interactions formed by the β-D-Gal moiety (Fig. 9), in agreement with the NMR data. Notably, the β-D-Gal moiety of Galacto-N-biose interacts with PPL residues with the same hydrogen bonding pattern as GalNAc, and also D-Gal observed in the chain B of the $P2_12_12$ structure of the PPL/D-Gal complex (Fig. 8h and Fig. 10). Details of the interactions between disaccharide and protein residues are reported in Table S1. Were Galacto-N-biose to bind PPL through the Gal moiety with the same binding mode as free galactose, steric clashes would result with Thr38 and Asn40. The Galacto-N-biose shows a slightly different conformation from the disaccharide model used for rigid-body modeling by Adachi et al. [12] which suggested TF antigen could not fit easily into the PPL binding site.

The STD NMR and crystallographic results reported here (using Gal, GalNAc and Galacto-N-biose as ligands) clarify the interactions formed by PPL with target sugars. NMR data revealed that PPL is able to recognize both anomers of D-GalNAc. In the X-ray structure, preferential binding of the α-anomer was observed, although the electron density suggests the presence of the β-anomer in the crystals as well. These results agree with the earlier crystal structure of this complex [12]. In the case of D-Gal, NMR showed that PPL exhibited exquisite preference for the β-anomer. D-Gal is reported to bind PPL about 40 times less strongly than D-GalNAc, for which the Kd is 0.2 mM [12]. The hexose ring of D-

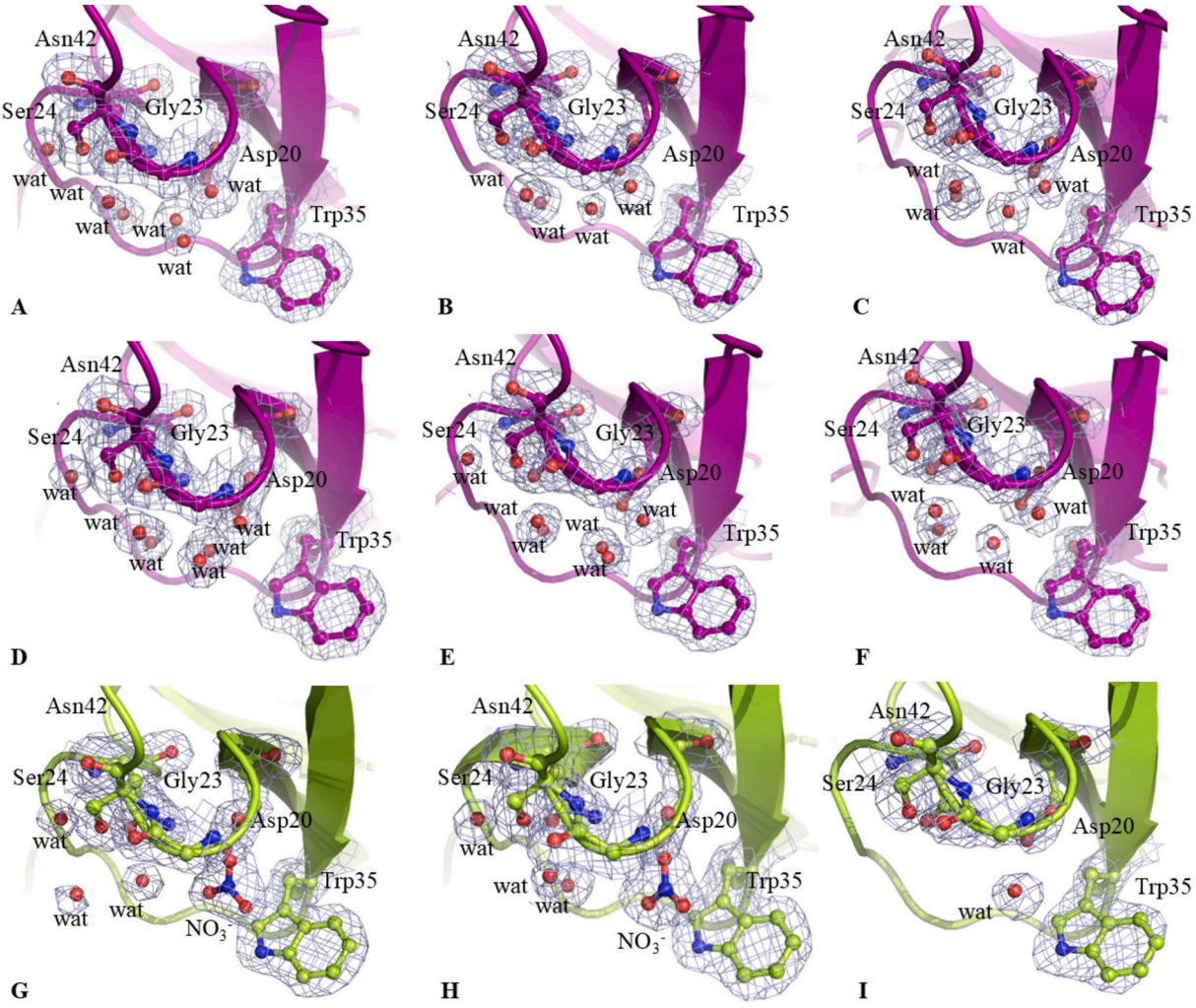


Fig. 7. The region corresponding to the D-GalNAc binding site in the two structures of ligand-free PPL ($P2_1$ structure in violet, $P2_12_12$ structure in green). 2Fo-Fc electron density maps are shown in light blue at 1.0 σ. Water molecules are shown as red spheres. In the chains A and B of the $P2_12_12$ structure, a nitrate ion occupies part of the D-GalNAc binding site.

Gal β-anomer forms similar interactions with PPL as D-GalNAc, lying against Trp35 and making hydrogen bonds between Asp20, Gly23, Asn42 and Ser24 and the O2/O3 and O4 atoms of the sugar (Figs. 4 B—D, Fig. 6A and C and Figs. 8 A-H). However, in almost every case the sugar ring of D-Gal is strongly rotated in the binding site relative to the binding mode of D-GalNAc, creating different hydrogen bonding pairs between the ligand and protein. Thus, D-Gal uses O2 and O3 to form hydrogen bonds with the Asp20 side-chain, while D-GalNAc uses O3 and O4 atoms. While O2 and O3 of galactose are equatorial, O4 is axial, so the geometry of hydrogen bonds formed by the Asp20 side chain is not equivalent in the two different sugar poses seen within the PPL binding pocket. The *N*-acetyl group of D-GalNAc contributes to ligand binding by forming an extra hydrogen bond with Ser24, but forbids the preferred binding mode of D-Gal, since this position of the sugar ring would cause a steric clash with Asn42 (Fig. 11). The O1 atom of GalNAc α-anomer sits 4.8 Å from the Nε1 atom of Trp35, too far for any steric clash. In contrast, the preferred binding mode of D-Gal α-anomer would bring O1 of the sugar much closer to the tryptophan side-chain, so only the β-anomer can bind. NMR and crystallographic analyses of Galacto-N-biose recognition by PPL showed that the galactose moiety of the disaccharide was accommodated in the sugar binding site, and the GalNAc oriented farther from the protein surface. The GalNAc residue of Galacto-N-biose is unable to bind PPL directly due to substitution at carbon-3, but the presence of the GalNAc residue forces the Gal moiety of the disaccharide to adopt an orientation in the binding site similar to that of monosaccharide GalNAc. Co-crystallization of PPL with Galacto-N-biose or D-Gal resulted in crystals of the same space-group as the apo form, another indicator of weak binding affinity.

5. Conclusions

In conclusion, we report here the molecular basis of glycan recognition by PPL, a selective R-type lectin that reads simple and minimal O-glycan units, by using a combination of NMR spectroscopy and X-ray crystallography.

Among the glycans analyzed here, α-linked GalNAc and Galβ(1–3) GalNAc constitute the saccharide moieties of Tn and TF, truncated O-glycans that occur very widely among cancers and are exploited as biomarkers and vaccine targets. Lectins with defined specificity for these motifs can serve as diagnostic tools or therapeutic guides. A PPL-based binder, engineered for safety and optimized affinity, could aid the detection of Tn- or TF-positive cells, or guide nanoparticles and imaging agents towards tumors. In diagnostics, labeled or immobilized PPL variants could enable selective detection of disease-associated O-glycans in tissues or biological fluids. Any translational use of PPL requires detailed knowledge of ligand poses within the binding site, and our integrated NMR/X-ray analysis provides the basis for developing PPL variants with modified selectivity or affinity, suitable for diagnostic or

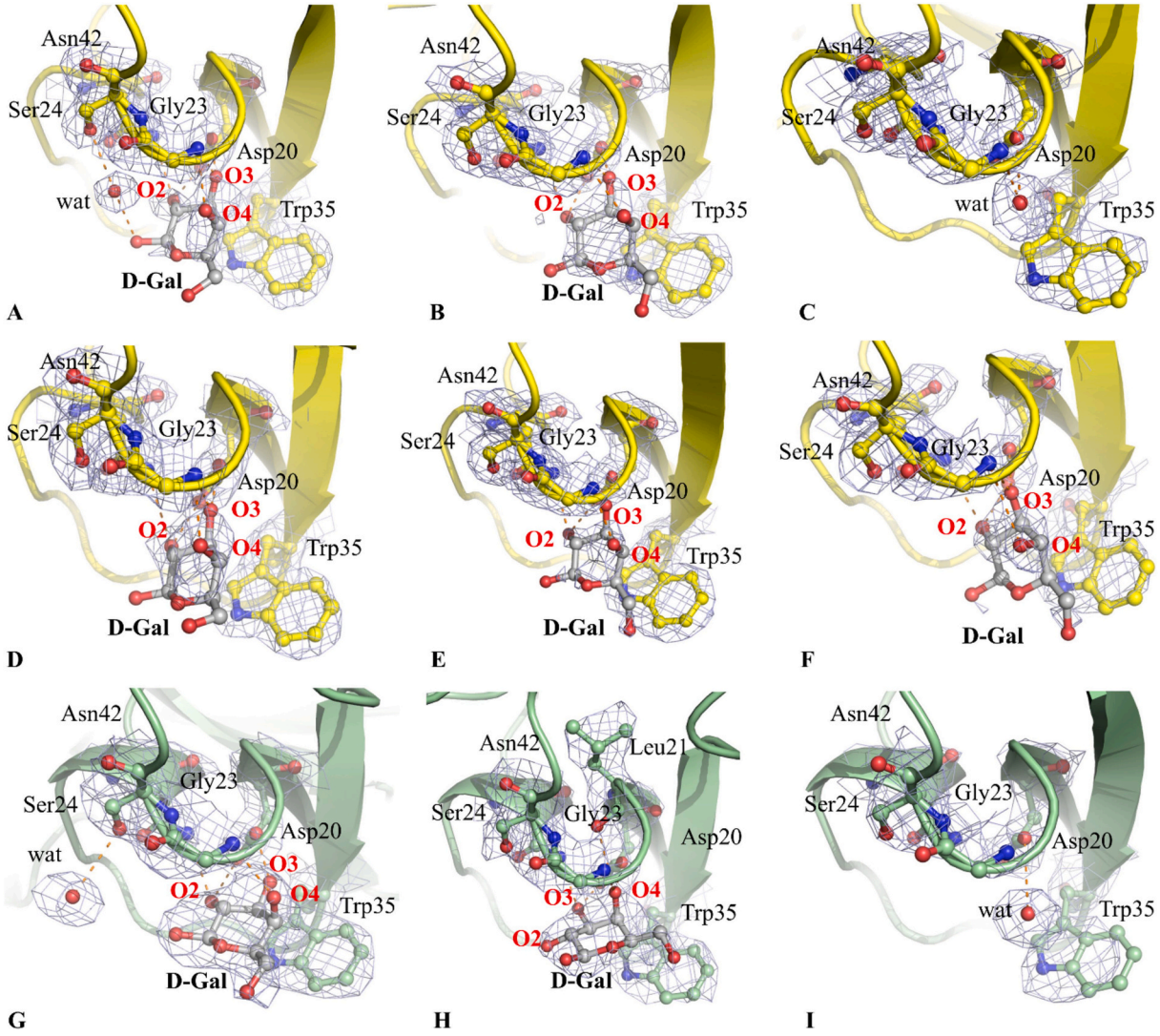


Fig. 8. D-Gal binding site in the two structures of PPL in complex with the sugar ($P2_1$ structure in yellow, $P2_12_12$ structure in light green). 2Fo-Fc electron density maps are shown in light blue at 1.0 σ. Water molecules are shown as red spheres. In the chains C of both $P2_1$ and $P2_12_12$ structures (panels C and I), the binding site is occupied by a water molecule. Hydrogen bonds are shown as orange dotted lines. In chain B of the $P2_12_12$ structure (panel H), D-Gal interacts with the protein via the O3 and O4 atoms, while in all the other chains, O2 and O3 atoms of D-Gal are involved in recognition by the protein. Ligand carbon atoms are in grey.

drug-targeting applications [34].

In particular, this study has solved several problems raised by the previous work of Adachi et al. [12] The altered binding mode of galactose could not have been predicted from the earlier X-ray model of the PPL:GalNAc complex, and the glycoarray study did not indicate TF antigen binding to PPL. This work highlights the advantages of NMR, especially STD, in analyses of weak sugar binding to lectins. Arrays are better suited to screens of widely varied ligands, but a combination of biophysical techniques has allowed us to characterize PPL ligand binding better than any single method could.

This integrated approach allowed us to observe details of interactions between PPL and three different ligands, D-Gal, D-GalNAc and Galacto-N-biose. NMR experiments demonstrated that PPL is able to recognize both anomers of the D-GalNAc sugar unit whereas it shows an exquisite selectivity for the β-anomer of D-Gal. Comparing the binding modes of D-GalNAc and D-Gal in the X-ray structures of their complexes with PPL a structural explanation for this difference has been found: D-Gal uses O2 and O3 atoms to interact with residues of the sugar binding site, while D-GalNAc uses O3 and O4 atoms and the *N*-acetyl group. The *N*-acetyl substituent of D-GalNAc alters its binding mode, preventing D-GalNAc from interacting with PPL in the same manner as D-Gal. These results were in accordance with NMR data, which showed that the protons at position 2, 3 and 4 of D-Gal were strongly involved in the binding with the protein, each exhibiting STD enhancements close to 100%. Moreover, the highest STD enhancement was observed for the proton at position 4 of α-D-GalNAc; also, the protons at position 3 and 5 showed significant STD effects (above 80%), indicating that this sugar region was strongly recognized by the protein.

The fully consistent NMR and X-ray results will help direct targeted engineering of PPL-based readers for cell-surface O-glycan detection, and by extension to other, longer, biologically relevant glycans [35]. The glycoarray data of Adachi et al. [12] indicated weak binding by H type saccharides (strongest for type III), and STD NMR will no doubt prove helpful in developing PPL variants with further reduced interactions with such ligands.

CRediT authorship contribution statement

Angela Marseglia: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Gabriella Tito:** Writing – review & editing, Visualization, Validation, Software, Methodology, Formal

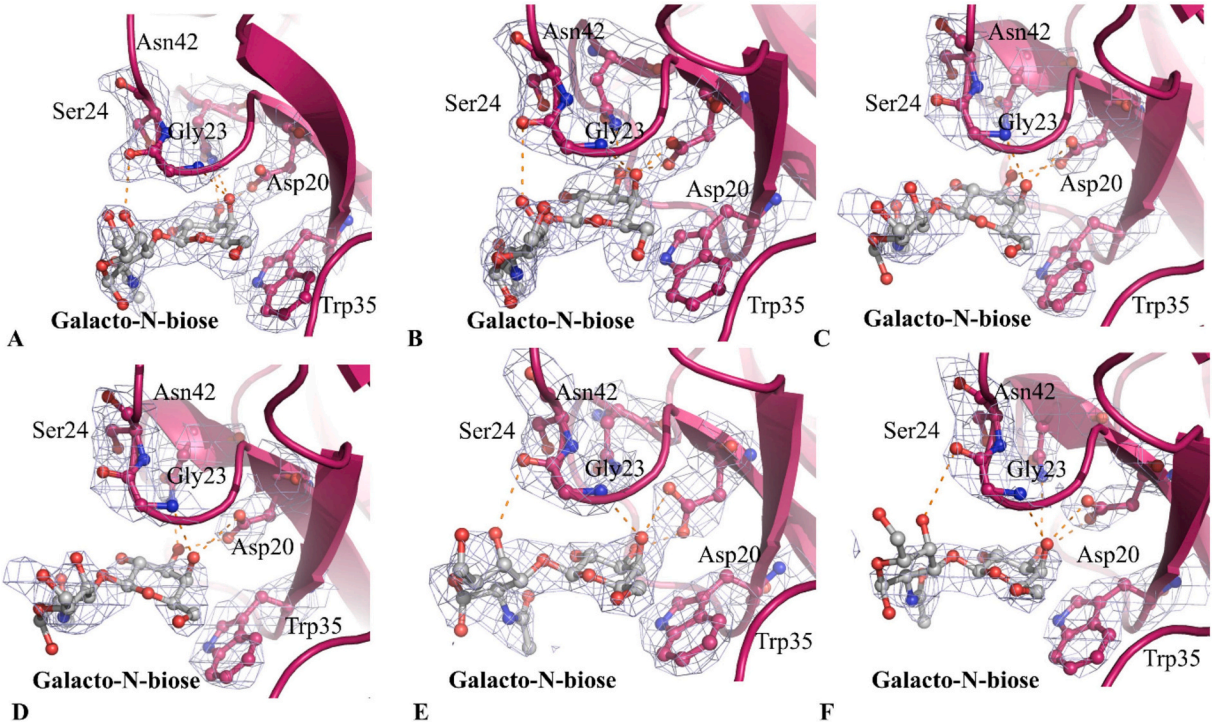


Fig. 9. Galacto-N-biose binding site in the structure of PPL in complex with the Galacto-N-biose. 2Fo-Fc electron density maps are reported in light blue at 1.0 σ. Ligand carbon atoms are in grey.

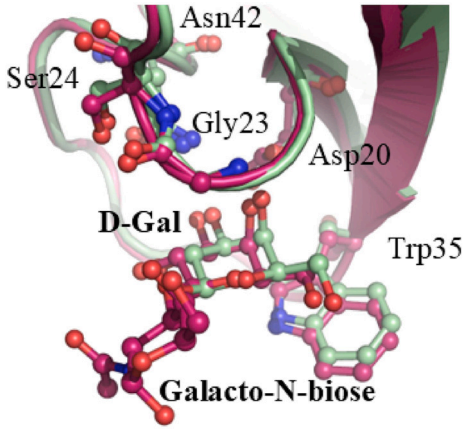


Fig. 10. Structural superimposition between the structures of PPL in complex with Galacto-N-biose (warm pink) and D-Gal (light green).

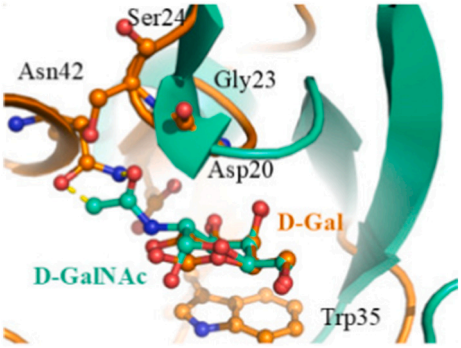


Fig. 11. Structural superimposition between D-GalNAc (green) and D-Gal (orange) highlighting the steric clash of the acetyl group of D-GalNAc with the side chain of Asn42, if GalNAc adopted this binding mode.

analysis, Data curation. **Giarita Ferraro:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis, Data curation. **Antonio Molinaro:** Writing – review & editing, Formal analysis. **Jeremy R.H. Tame:** Writing – review & editing, Formal analysis. **Kenichi Kamata:** Writing – review & editing, Formal analysis. **Antonello Merlino:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Investigation, Formal analysis, Data curation. **Roberta Marchetti:** Writing – review & editing, Writing – original draft, Validation, Supervision, Investigation, Formal analysis, Data curation.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijbiomac.2026.153685.

Data availability

Data will be made available on request.

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